

SAI VENKATA HARSHINI RAYAVARAPU

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SUMMARY

Machine Learning and Data Engineer with 14 months of production data engineering experience at Accenture and an MS in Computer Science from Cleveland State University. Builds end-to-end systems: two-tower neural retrieval serving sub-50ms p95 latency, multi-agent LLM pipelines with automated SQL validation, and causal inference frameworks that separate true treatment effect from correlation. Deep SQL performance background across high-volume financial data.

TECHNICAL SKILLS

- **Languages:** Python (PyTorch, Pandas, NumPy, Scikit-Learn), SQL, C++, COBOL / JCL
- **Machine Learning:** Two-Tower Retrieval, Embeddings & ANN Search, Transformers, LLMs (Llama 3.3), Agentic RAG, Causal Inference, Propensity Score Matching, A/B Testing, NLP (TF-IDF)
- **Data & Infrastructure:** FAISS, FastAPI, Docker, DuckDB, MLflow, Git, Power BI, Streamlit
- **Practices:** MLOps, Multi-Agent Orchestration, Counterfactual Analysis, Query Optimization, System Design

PROJECTS

NeuralStream - Neural Retrieval & Ranking System [\[GitHub\]](#) PyTorch | FAISS | FastAPI | Docker

- Built a two-stage retrieval-and-ranking pipeline: a two-tower network embeds users and items in a shared 64-D space.
- Served approximate nearest-neighbor search with FAISS, holding recall@10 while sustaining sub-50ms p95 query latency across a 1M-item catalog on a single node.
- Shipped the ranker as a containerized FastAPI service with a live dashboard exposing normalized match confidence.

Starbit Engine - Autonomous Multi-Agent Data Intelligence [\[GitHub\]](#) LangGraph | Llama 3.3 | DuckDB

- Architected a LangGraph multi-agent system that plans, writes, validates, and visualizes SQL against DuckDB without human intervention.
- Added a critic agent that catches hallucinated or invalid SQL and triggers bounded recursive retries, cutting failed queries by 85% on a held-out benchmark of multi-table schemas.
- Instrumented an animated data-lineage view in ECharts for step-level observability from raw ingestion to final result.

Causal Analytics - Incremental Lift & ROI Framework [\[GitHub\]](#) Scikit-Learn | Statsmodels | Pandas

- Built a causal inference pipeline measuring true incremental lift of product features, exposing a 7x over-estimation in the naive correlational estimate.
- Constructed counterfactual control groups via Propensity Score Matching (KNN), isolating treatment effect from self-selection bias in high-engagement user segments.
- Recovered known ground-truth ROI within 4.5% on a simulated campaign dataset, validating the estimator before applying it to observational data.

PROFESSIONAL EXPERIENCE

Accenture Pvt Ltd Bengaluru, India

Application Development Associate / Software Engineering Analyst October 2022 - December 2023

- Optimized high-volume banking data pipelines in Mainframe SQL, refactoring execution plans and indexing strategies to cut data retrieval latency by 15%.
- Re-engineered legacy reporting into modern SQL with automated audit controls, ending recurring data-corruption incidents.
- Replaced manual banking reconciliation workflows with automated SQL-based analytical pipelines, removing over 12 hours of manual processing per week.
- Authored data governance and system maintenance documentation that passed external regulatory audits against banking security and SOC 2 controls.

EDUCATION

Cleveland State University Cleveland, OH

Master of Science in Computer Science | GPA: 3.4 / 4.0 January 2024 - May 2026

- **Coursework:** Artificial Intelligence, Machine Learning, Data Science, Network Security & Privacy.

Vignan's Institute of Information Technology Visakhapatnam, India

Bachelor of Technology in Computer Science & Engineering | GPA: 8.39 / 10.0 May 2022

CERTIFICATIONS

- Databricks Accreditation - AI Agent Fundamentals & Databricks Fundamentals (2026) | AWS - Fundamentals of ML & AI (2026)

Work Authorization: F-1 STEM OPT Eligible | 36 Months Immediate Work Authorization (No Initial Sponsorship Required)